Seminar 6 - Warm-up

6.1. Sum-of angles-formulas for sine/cosine using rotation matrix.

- A rotation with angle θ followed by a rotation with angle $\tilde{\theta}$ is the same as a rotation with angle $\theta + \tilde{\theta}$.

$$\boxed{Rot_{\tilde{\theta}} \circ Rot_{\theta} = Rot_{\tilde{\theta}+\theta}}$$

$$\begin{bmatrix} \cos\tilde{\theta} & -\sin\tilde{\theta} \\ \sin\tilde{\theta} & \cos\tilde{\theta} \end{bmatrix} \cdot \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} \cos(\tilde{\theta}+\theta) & -\sin(\tilde{\theta}+\theta) \\ \sin(\tilde{\theta}+\theta) & \cos(\tilde{\theta}+\theta) \end{bmatrix}$$

$$= \begin{bmatrix} \cos\tilde{\theta}\cos\theta - \sin\tilde{\theta}\sin\theta & -(\cos\tilde{\theta}\sin\theta + \sin\tilde{\theta}\cos\theta) \\ \dots & \dots \end{bmatrix}$$

6.2. Matrix form of an affine map obtained from a rotation with $\frac{\pi}{3}$ after a translation with $(-1,0)$.

$$\Phi(P) = Rot_{\frac{\pi}{3}} \circ T_{(-1,0)}(P) = Rot_{\frac{\pi}{3}} \cdot \left(P + \begin{bmatrix} -1 \\ 0 \end{bmatrix}\right)$$

$$= \begin{bmatrix} \cos\frac{\pi}{3} & -\sin\frac{\pi}{3} \\ \sin\frac{\pi}{3} & \cos\frac{\pi}{3} \end{bmatrix} \cdot \left(P + \begin{bmatrix} -1 \\ 0 \end{bmatrix}\right)$$

$$= \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \cdot \left(P + \begin{bmatrix} -1 \\ 0 \end{bmatrix}\right) \qquad 2\times\cancel{2}\ \cancel{2}\times 1$$

$$= \frac{1}{2} \cdot \begin{bmatrix} 1 & -\sqrt{3} \\ -\sqrt{3} & 1 \end{bmatrix} P + \frac{1}{2} \cdot \begin{bmatrix} 1 & -\sqrt{3} \\ -\sqrt{3} & 1 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{2} \cdot \begin{bmatrix} 1 & -\sqrt{3} \\ -\sqrt{3} & 1 \end{bmatrix} P + \frac{1}{2} \cdot \begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix}.$$

6.3. Verify that the matrices belong to SO(2). → rotation angle?

O(m) → set of orthogonal matrices. ($A^TA = I_m$) => $\det(A) = \pm 1$.

if $\det(A) = 1$ => SO(m) → special orthogonal.

Check: change sign => O(n), not SO(n).

A: $A = \frac{1}{5}\begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix}$; ~~$\det A = \frac{1}{5}(16+9) = 5$?~~

Method I: $\underbrace{A \cdot A^T}_{} = \frac{1}{5}\overbrace{\begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix}}^{A} \cdot \frac{1}{5}\overbrace{\begin{bmatrix} 4 & 3 \\ -3 & 4 \end{bmatrix}}^{A^t} = \frac{1}{25} \cdot \begin{bmatrix} 25 & 0 \\ 0 & 25 \end{bmatrix} = I_2.$

=> $A \in O(2)$, dimension 2.
$\det A = 1$
} => $A \in SO(2)$.

Method II: $\left(\frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2 = 1$ => $\exists \theta$ s.t. $\begin{cases} \frac{4}{5} = \cos\theta \\ \frac{3}{5} = \sin\theta \\ A = Rot_\theta. \end{cases}$

B: B is the matrix $Rot_{90°}$.

$B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \cos\frac{\pi}{2} & -\sin\frac{\pi}{2} \\ \sin\frac{\pi}{2} & \cos\frac{\pi}{2} \end{bmatrix}$; $\det B = 1$ => SO(2).

C: $C = \frac{1}{13}\begin{bmatrix} 5 & -12 \\ 12 & 5 \end{bmatrix}$

$\det C = \left(\frac{5}{13}\right)^2 + \left(\frac{12}{13}\right)^2 = \frac{25+144}{169} = 1.$ => SO(2)

D: $D = \frac{1}{17}\begin{bmatrix} 8 & 15 \\ -15 & 8 \end{bmatrix}$

$\det D = \left(\frac{8}{17}\right)^2 + \left(\frac{15}{17}\right)^2 = \dots = 1.$ => SO(2).

=> In all cases, changing the sign of a column or row multiplies the det. with -1.

?! $A = \frac{1}{5}\begin{bmatrix} -4 & 3 \\ 3 & -4 \end{bmatrix} = \left(\frac{+4}{5}\right)^2 - \left(\frac{3}{5}\right)^2 = \frac{16-9}{25}$?!

6.4. Change column sign (last column) $\Rightarrow$ orthogonal reflection. ?

$\Rightarrow$ reflection lines

$$A = \frac{1}{5}\begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix}, \quad \tilde{A} = \frac{1}{5}\begin{bmatrix} 4 & 3 \\ 3 & -4 \end{bmatrix}$$

For all cases $\Rightarrow$ reflection line of the reflection $\begin{bmatrix} x \\ y \end{bmatrix} \mapsto \tilde{A}\begin{bmatrix} x \\ y \end{bmatrix}$ is the set of points which don't move under the reflection.

?? $\Rightarrow$ the set of solutions to $\tilde{A}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \Leftrightarrow \begin{bmatrix} -1 & 3 \\ 3 & -9 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\Leftrightarrow x = 3y. \Rightarrow$ the line is $x - 3y = 0$ or $\begin{cases} x = 3t \\ y = t \end{cases}$

test: $\frac{1}{5}\begin{bmatrix} 4 & 3 \\ 3 & -4 \end{bmatrix}\begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

?? $\Rightarrow$ direction vectors for the line = eigenvectors for eigenvalue 1.

6.5. Check that matrices belong to $SO(3)$.

$$\boxed{A \in SO(3) \Leftrightarrow AA^T = I_3 \ \& \ \det A = 1.}$$

$$A = \frac{1}{3}\begin{bmatrix} -1 & 2 & -2 \\ -2 & -2 & -1 \\ -2 & 1 & 2 \end{bmatrix}$$

$$A \cdot A^T = \frac{1}{3}\begin{bmatrix} -1 & 2 & -2 \\ -2 & -2 & -1 \\ -2 & 1 & 2 \end{bmatrix} \cdot \frac{1}{3}\begin{bmatrix} -1 & -2 & -2 \\ 2 & -2 & 1 \\ -2 & -1 & 2 \end{bmatrix}$$

$$= \frac{1}{9}\begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3.$$

$\det(A \cdot A^T) = \det I_3 = 1. \Rightarrow A \in SO(3)$.

6.6. $\gamma(t) = (t - \sin(t), 1 - \cos(t)) \rightarrow$ by rolling a circle of radius 1 on the x-axis.

~ <u>Cycloids</u> : $\mathbb{E}^2$.

Homog. matrix of a rotation around $C(c_1, c_2) \in \mathbb{E}^2$:

$$\begin{bmatrix} 1 & 0 & c_1 \\ 0 & 1 & c_2 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -c_1 \\ 0 & 1 & -c_2 \\ 0 & 0 & 1 \end{bmatrix} =$$

$$= \begin{bmatrix} \cos\theta & -\sin\theta & c_1 \\ \sin\theta & \cos\theta & c_2 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -c_1 \\ 0 & 1 & -c_2 \\ 0 & 0 & 1 \end{bmatrix} = \cancel{\begin{bmatrix} \cos\theta & -\sin\theta \end{bmatrix}}$$

$$= \begin{bmatrix} \cos\theta & -\sin\theta & -c_1 \cdot \cos\theta + c_2 \cdot \sin\theta + c_1 \\ \sin\theta & \cos\theta & -c_1 \sin\theta - c_2 \cos\theta + c_2 \\ 0 & 0 & 1 \end{bmatrix}$$

center $C \to C(0,1) \Rightarrow$ homog. rotation matrix:

$$\begin{bmatrix} \cos\theta & -\sin\theta & \sin\theta \\ \sin\theta & \cos\theta & -\cos\theta + 1 \\ 0 & 0 & 1 \end{bmatrix} \cdot \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{\text{homog. coord. of the origin}} = \begin{bmatrix} \sin\theta \\ -\cos\theta + 1 \\ 1 \end{bmatrix}$$

Vary θ with time t $\Rightarrow$ rotate origin around C <u>counterclockwise</u>.

Change sign of angle $\Rightarrow$ clockwise rotation:

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \longmapsto \begin{bmatrix} -\sin(t) \\ -\cos(t) + 1 \\ 1 \end{bmatrix}$$

At the same time (t) $\Rightarrow$ translate the points along the x-axis, in direction of i:

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \longmapsto \begin{bmatrix} 1 & 0 & t \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -\sin t \\ -\cos t + 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -\sin t + 1 \\ -\cos t + 1 \\ 1 \end{bmatrix}$$

$\Rightarrow$ cycloid.